

Splunk

🔍 Investigation Report: Widget LLC - Suspicious Activity on Finance Department Endpoint

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Client: Widget LLC

Platform: Splunk

Objective: Identify suspicious or malicious activity on a Financial Analyst's endpoint that had endpoint protection disabled during the specified period.

🛠 Tools Used

- Splunk

🔍 Investigation Steps

1. A Web Browser Password Viewer is executed on the infected machine. What is the name of the binary? Enter the full path.

The screenshot shows a Splunk search interface with the query 'password viewer' and a time range of 'All time'. The search results show 27 events. The 'Events (27)' tab is selected, displaying a timeline view with green bars representing events. Below the timeline, a table lists the events. The first event is highlighted, showing details for a 'Web Browser Password Viewer' executed on 12/29/21 at 2:09:50.000 AM. The event details include the full path of the binary: C:\Users\FINANC~1\AppData\Local\Temp\11111.exe.

1 password viewer All time

✓ 27 events (before 4/21/25 12:56:30.000 AM) No Event Sampling Job II Smart Mode

Events (27) Patterns Statistics Visualization

Format Timeline Zoom Out Zoom to Selection Deselect 1 minute

List Format 20 Per Page Prev 1 2

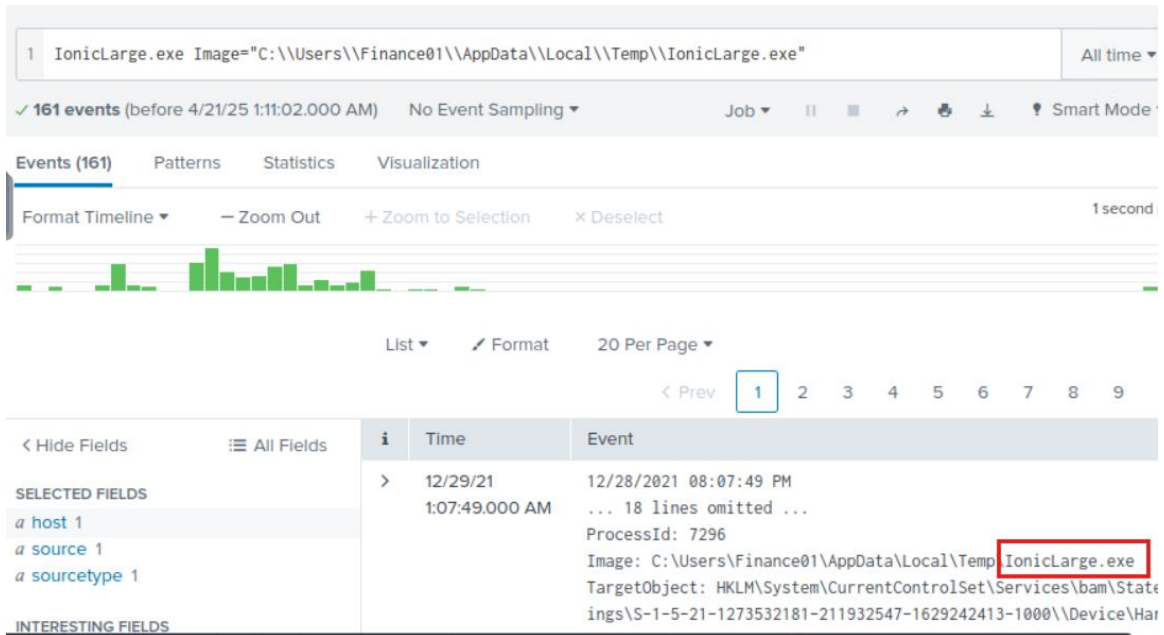
	i	Time	Event
>		12/29/21 2:09:50.000 AM	... 20 lines omitted ... ImageLoaded: C:\Users\FINANC~1\AppData\Local\Temp\11111.exe FileVersion: 2.06 Description: Web Browser Password Viewer Product: - Company: NirSoft Show all 31 lines host = DESKTOP-H1ATIJC

< Hide Fields All Fields

SELECTED FIELDS
a host 1
a source 1
a sourcetype 1

INTERESTING FIELDS
a CommandLine 1
a Company 1

2. Another suspicious binary running from the same folder was executed on the workstation. What was the name of the binary? What is listed as its original filename?



```
source = WinEventLog:Microsoft-Windows-Sysmon/Operational
sourcetype = WinEventLog:Microsoft-Windows-Sysmon/Operational

> 12/29/21      ... 19 lines omitted ...
1:06:32.000 AM Image: C:\Users\Finance01\AppData\Local\Temp\IonicLarge.exe
FileVersion: 3.22.19.1
... 3 lines omitted ...
OriginalFileName: PalitExplorer.exe
CommandLine: "C:\Users\Finance01\AppData\Local\Temp\IonicLarge.exe"
CurrentDirectory: C:\Users\Finance01\AppData\Local\Temp\
Show all 38 lines
host = DESKTOP-H1ATIJC
```

3. The binary from the previous question made two outbound connections to a malicious IP address. What was the IP address?

DestinationIp

8 Values, 54.037% of events

Selected

Reports

Top values Top values by time Rare values

Events with this field

Values	Count	%
127.0.0.1	79	90.804%
2.56.59.42	2	2.299%
104.27.40.48	1	1.149%
142.250.191.132	1	1.149%
142.250.191.206	1	1.149%
148.251.234.93	1	1.149%

4. The same binary made a change to a registry key. What was the key path?

< Hide Fields All Fields List Format 20 Per Page

Time Event 1 2 3 4 5 6 7 8 9 Next >

Details: DWORD (0x00000001)

Show all 24 lines

host = DESKTOP-H1ATIJC

source = WinEventLog:Microsoft-Windows-Sysmon/Operational

sourcetype = WinEventLog:Microsoft-Windows-Sysmon/Operational

> 12/29/21 1:06:38.000 AM

12/28/2021 08:06:38 PM

... 18 lines omitted ...

ProcessId: 7296

Image: C:\Users\Finance01\AppData\Local\Temp\IonicLarge.exe

TargetObject: HKLM\SOFTWARE\Policies\Microsoft\Windows Defender\Real-Time Protection\DisableScanOnRealtimeEnable

Details: DWORD (0x00000001)

Show all 24 lines

host = DESKTOP-H1ATIJC

source = WinEventLog:Microsoft-Windows-Sysmon/Operational

5. The attacker ran several commands within a PowerShell session to change the behaviour of Windows Defender. What was the last command executed in the series of similar commands?

Got after using the CommandLine option after searching for powershell

The screenshot shows the Windows Event Viewer interface. The top bar indicates '1 event (before 4/21/25 1:29:50.000 AM)' and 'No Event Sampling'. The 'Events (1)' tab is selected. The event list shows a single event with the following details:

Time	Event
12/29/21 1:07:52.000 AM	12/28/2021 08:07:52 PM ... 22 lines omitted ... Company: Microsoft Corporation OriginalFileName: Cmd.Exe CommandLine: "C:\Windows\System32\cmd.exe" /C forfiles /p c:\w 32 /m waitfor.exe /c "cmd /C powershell WMIC /NAMESPACE:\\root ndows\Defender PATH MSFT_MpPreference call Add ThreatIDDefault 47735503 ThreatIDDefaultAction_Actions=6 Force=True" & forfile ws\system32 /m where.exe /c "cmd /C powershell WMIC /NAMESPACE soft\Windows\Defender PATH MSFT_MpPreference call Add ThreatID Ide=2147737010 ThreatIDDefaultAction_Actions=6 Force=True" &

The 'Selected Fields' list includes: CommandLine 1, host 1, source 1, and sourcetype 1. The 'Interesting Fields' list includes: Company 1, ComputerName 1, and CurrentDirectory 1.

6. Based on the previous answer, what were the four IDs set by the attacker? Enter the answer in order of execution.

The screenshot shows a PowerShell command execution event. The command line is as follows:

```

... 22 lines omitted ...
Company: Microsoft Corporation
OriginalFileName: Cmd.Exe
CommandLine: "C:\Windows\System32\cmd.exe" /C forfiles /p c:\windows\system
32 /m waitfor.exe /c "cmd /C powershell WMIC /NAMESPACE:\\root\Microsoft\Wi
ndows\Defender PATH MSFT_MpPreference call Add ThreatIDDefaultAction_Ids=21
47735503 ThreatIDDefaultAction_Actions=6 Force=True" & forfiles /p c:\windo
ws\system32 /m where.exe /c "cmd /C powershell WMIC /NAMESPACE:\\root\Micro
soft\Windows\Defender PATH MSFT_MpPreference call Add ThreatIDDefaultAction
_Ids=2147737010 ThreatIDDefaultAction_Actions=6 Force=True" & forfiles /p
c:\windows\system32 /m calc.exe /c "cmd /C powershell WMIC /NAMESPACE:\\roo
t\Microsoft\Windows\Defender PATH MSFT_MpPreference call Add ThreatIDDefaul
tAction_Ids=2147737007 ThreatIDDefaultAction_Actions=6 Force=True" & forfil
es /p c:\windows\system32 /m notepad.exe /c "cmd /C powershell WMIC /NAMESP
ACE:\\root\Microsoft\Windows\Defender PATH MSFT_MpPreference call Add Threa
tIDDefaultAction_Ids=2147737394 ThreatIDDefaultAction_Actions=6 Force=True"
&
CurrentDirectory: Add to search
Show all 38 lines
Exclude from search

```

The four IDs are: 2147735503, 2147737010, 2147737007, and 2147737394.

7. Another malicious binary was executed on the infected workstation from another AppData location. What was the full path to the binary?

1 AppData All time

8,598 events (before 4/21/25 1:35:55.000 AM) No Event Sampling Job

Events (8,598) Patterns Statistics Visualization

Format Timeline Zoom Out Zoom to Selection Deselect 1 day per

List Format 20 Per Page

1 2 3 4 5 6 7 8 ... Ne

< Hide Fields All Fields

SELECTED FIELDS

- CommandLine 62
- host 1
- source 3
- sourcetype 3

Time Event

12/29/21 12/28/2021 09:11:28 PM

2:11:28.000 AM ... 17 lines omitted ...

ProcessId: 12576

Image: C:\Users\Finance01\AppData\Roaming\EasyCalc\EasyCalc.exe

TargetFilename: C:\Users\Finance01\AppData\Local\EasyCalc\User D

8. What were the DLLs that were loaded from the binary of the previous question? Enter the answers in alphabetical order.

Used the ImageLoaded option to check the DLL

< Hide Fields All Fields

SELECTED FIELDS

- CommandLine 62
- host 1
- ImageLoaded 74
- source 3
- sourcetype 3

INTERESTING FIELDS

- CallTrace 100+
- ComputerName 2
- CreationUtcTime 100+
- EventCode 25
- EventType 7
- GrantedAccess 8
- Image 59
- Index 1

Top 10 Values

Value	Count	%
C:\Users\FINANC~1\AppData\Local\Temp\11111.exe	23	11.274%
C:\Users\Finance01\AppData\Roaming\EasyCalc\EasyCalc.exe	15	7.353%
C:\Users\Finance01\AppData\Roaming\EasyCalc\new_elf.dll	15	7.353%
C:\Users\Finance01\AppData\Roaming\EasyCalc\ffmpeg.dll	13	6.372%
C:\Users\Finance01\AppData\Roaming\EasyCalc\new.dll	13	6.372%
C:\Windows\SysWOW64\urlmon.dll	13	6.372%
C:\Users\Finance01\AppData\Local\Programs\Setup.exe	6	2.941%

🌱 **Final Thoughts:**

The forensic investigation into Widget LLC's Financial Analyst endpoint revealed multiple Indicators of Compromise (IOCs) and post-exploitation activities consistent with credential theft, system tampering, and potential command-and-control (C2) communication. The attacker leveraged tools like NirSoft's browser password viewer, disguised binaries, PowerShell-based tampering of Windows Defender, and persistence techniques through registry manipulation.

✅ **Recommendations to Prevent Insider Data Exfiltration**

- **Immediate Host Isolation:** Quarantine the affected system to prevent lateral movement and further data exfiltration.
- **Credential Reset:** Invalidate all credentials used on the compromised host, especially those stored in browsers.
- **Full Endpoint Reimage:** Due to deep compromise, a complete OS reinstallation is advised to eliminate any remaining backdoors or malware.
- **Network-wide IOC Sweep:** Search all endpoints and servers for matching filenames, registry changes, and IP connections (2[.]56[.]59[.]42) to identify potential spread.
- **Implement EDR/XDR:** Deploy robust endpoint detection and response tools across all Finance Department endpoints.
- **Review Security Controls:** Reinforce Group Policy to prevent execution from AppData and ensure Defender settings cannot be altered by users or scripts